Inland ENC Harmonization Group

27 - 28 October 2026
online

Welcome, introduction, organization

Introductions of participants (please keep it short)

Organizational details**:**

All documents available at
https://github.com/IEHG/IEHG_meeting_2026

please send your presentations (if any) to bernd.birklhuber@bmimi.gv.at)

All presentations will be published at https://ienc.openecdis.org
(please inform, if you object)

Agenda (1)

1. Welcome, introduction of participants, organizational details
2. Update on the legal and organizational background and the status of implementation
3. Presentation of Inland ENC applications by private companies
4. Election of core group, ToR and list of members
5. Test charts for edition 2.6
6. Status of S-99, S-100, S-101 and other S-100 based products
7. Interoperability and S-98, S-403
8. Development of the S-401 main document
9. Development of the S-401 DCEG
10. Development of the S-401 Feature Catalogue
11. Development of the S-401 Portrayal Catalogue
12. Development of S-401 test cells
13. Development of a S-401 alert catalogue
14. Development of S-158:401 validation checks
15. S-57 to S-100 converter and conversion guidance
16. S-401 and S-403 conclusions and timeline

Agenda

Agenda (2)

17. S-402, S-201 and S-125
18. Annual report to HSSC
19. Information documents on Inland ENCs
20. Future operation of ienc website and discussion forum
21. Any other business
22. Next meeting

Most important goals of this meeting are

- the development of the first operational edition 2.0.0 of S-401
- and an agreement on the content of S-403
- and a decision whether we will develop S-402 or use S-102 and S-104

Time schedule

Tuesday, 27 October and Wednesday, 28 October

- 12:00 UTC start of the meeting
- 15:00 UTC latest end of the meeting

Inland ENCs and the Inland ENC Harmonization Group (IEHG)

Inland ENCs

Update on the legal and organizational background and the status of implementation

Brazil

China

Europe

Russian Federation

USA

- Only important news, ≤ 5 min

Overview

Inland ENC Harmonization Group

Open action points

- Please provide an email to bernd.birklhuber@bmimi.gv.at containing updates to the information on Inland ENC coverage [offline version]
- Please check also the list of international and national organizations on the website and inform bernd.birklhuber@bmimi.gv.at about necessary updates

Inland ENC applications

- Caris
- Esri
- IIC Technologies
- Navico
- Periskal

List of IENC applications:
https://ienc.openecdis.org/links
[offline version]

Only important news, ≤ 5 min

Open action points

- The IEHG website contains also a list of both National Organizations and Private Companies involved with IENCs
  [offline version]

  If not listed, please provide URL to bernd.birklhuber@bmimi.gv.at for inclusion on the website

Election of the Core Group

- By simple majority vote, chairpersons, vice-chairs and technical coordinators are elected.
  - Chair – Two persons (co-chairs) each from a different region. Only representatives of waterway authorities can become chairpersons.
  - One vice-chair from each region, which is not already a chair.
  - Technical Coordinators - One technical coordinator for each region.
  - Core Group – The two Chairs, the vice-chairs and Technical Coordinators.
  - The S-100 representatives are also part of the Core Group

ToR

Election of chairs

- Chair – Two persons (co-chairs) each from a different region. Only representatives of waterway authorities can become chairpersons.
- One vice-chair from each region, which is not already a chair

Current co-chairs:

Denise LaDue (USACE, North America)
Bernd Birklhuber (BMIMI, Europe)

Current vice-chairs:

Andre Alves Araujo, (DHN, South America)
Qun Gu (WTI, Asia)

Proposals for the coming period:

Gerald Thornberry (USACE) is proposed to replace Denise









Inland ENC
Harmonization Group










Two Laughing Men


DID YOU CALL BRIGADE?

- Thank you, Denise, for your commitment for such a long time!
- Thank you for serving as Co-chair for 18 years (since 2008)!
- It has really been a pleasure to work with you
- IEHG and the Product Specification for IENCs wouldn’t be the same without you
- We hope to see you at the next meeting to say good bye in person

Election of technical coordinators

- Technical Coordinators - One technical coordinator for each region

  Current technical coordinators:

  Cameron McLeay (CARIS USA, North America)
  Gaël Billet (Periskal Group, Europe)
  Vladimir Sekachev (STEOR-NSB LLC, Russia)
  ?? (South America)
  Li Liu (WTI, Asia)

  Proposals for the coming period?

Election of representatives for S-100

IEHG has to nominate representatives for the registry management of S-100:

- representatives as submitting organization
  (currently Denise LaDue and Gaël Billet)
- representatives for the Domain Control Body
  (currently Denise LaDue and Gert Morlion)
- representatives for the Executive Control Body
  (currently Denise LaDue and Gaël Billet)
- Proposals for the coming period?
  Gerald Thornberry to replace Denise LaDue

ToR, Annex B

Amendment of the ToR

IEHG could check whether the ToR of IEHG need to be updated

ToR

Update of the Introduction of the EG

- The Introduction to the DCEG has been copied from the EG 2.6
- It has to be amended to cover the transition to S-401
- Proposed way forward:
  - develop a proposal for the amendment with open dates and submit it as a CR
  - add the dates after the adoption of the first operational edition

List of members and participants

- The list is based on the ToR and is containing
  - Members
  - Expert contributors
  - Participants

- Some e-mail addresses are no longer active and we can’t contact those persons
- Please check if those persons can be deleted or if you know their new addresses

List of members

Test charts for edition 2.6

- The Recommended Validation Checks for Inland ENCs edition 2.6.1 (based on Special Publication S-58 Ed. 8.0.0 of IHO) have been adopted and published
- Action point from last meeting:
  Denise to develop test charts for edition 2.6 based on the test charts for edition 2.5
- Status?
- 3 test charts are available, but without a reference of the correct presentation
  Do we need one or is the PresLib sufficient?

Status of S-99, S-100, S-101

- S-100 has entered into force on 1st January 2010
- S-100 edition 5.2.1 has been published in December 2025
- S-100 Registry 3.1
- S-99 edition 2.0.0 has been published in October 2022
- S-101 edition 2.0.0 has been published in January 2025
- S-101 draft edition 2.1.0 has been published in July 2026

  for adoption by HSSC, but will be amended again (goal: end of 2026)
- All adopted documents available at https://iho.int/en/standards-and-specifications

S-101 dual fuel period

- The dual fuel period for S-101 has started on 01 01 2026
- Chart producers are invited to start the production and publication of S-101 ENCs
- There is currently no deadline for the provision of a full coverage with S-101
- S-57 ENCs have to be published until the end of the dual fuel period (not fixed)
- No type approved dual fuel ECDIS available
- Users of dual fuel ECDIS will be free to use either S-57 or S-101 ENCs
- Currently no end date for the equipment with S-100 based (dual fuel) ECDIS

Status of S-99, S-100, S-101

Action points from last meeting (1):

- keep track of S-101 developments and the consequences for IENCs (Cameron, Tom, Friedhelm, Gert, Gaël)

- Denise to reach out to Tom Bovey regarding S-501 developments to determine whether there will be similar challenges faced by S-501 regarding Interoperability with S-101. Interoperability will be an issue for S-501 also

Status of S-99, S-100, S-101

Action points from last meeting (2):

- Check the new edition of S-52

  A:100, IHO S-100 ECDIS Presentation Library for S-57 ENC Edition 5.0(.0), has been published in October 2025, but is only made available to registered OEMs
- The domain managers to mail to Jeff and to make him aware of duplications due to violations of S-99
- Clarify if a registry access is necessary for the registration of a new producer code

Application for new Producer Codes

- https://registry.iho.int/producercode/list2.do contains new Producer Codes for S-100 based Products
- New producer codes will be necessary for the production of S-401 IENCs

Interoperability and S-98

- IEHG decided to develop an own interoperability standard
- IHO has reserved the PS number S-403 for the inland interoperability standard
- We decided that S-403 should be based on S-98 and should only contain S-401 and either S-402 or S-102 with S-104
- S-98 is now not only containing the interoperability, but also the minimum requirements for ECDIS (e.g. screen dimensions and colors) and functionality (e.g. route planning, boatmasters entries, manual update FC, depth display, …)
- Up to now IEHG had no minimum requirements for Inland ECDIS or ECS
- Do we include the (adapted) ECDIS requirements of S-98 in S-403 or should the Inland ECDIS/ECS requirements be defined by the regions?

Interoperability and S-98 (2)

- S-98 is only applicable to type approved ECDIS, not to ECS
- If we include minimum requirements in S-403, for which applications should they be valid?
- In Europe we distinguish between
  - navigation mode (type approved, with radar overlay)
  - information mode (often just software, installed by the user on a PC, laptop, tablet)

S-403 content

Should S-403

- Only contain interoperability?
- Also contain minimum requirements for Inland ECDIS and Inland ECS?
- Also contain requirements for type approval? Would a type approval of one country be recognized by all other regions?
- Should it be possible to sell just software?
- Should hardware (especially display) requirements (size, colors, …) apply to all?
- Do we include interoperability levels 1 and 2 or only one of them?

Interoperability and S-98, action points (1)

- Check if S-101 ENCs and S-401 IENCs (without overlap) are displayed correctly when the vessel moves from one area to the other without an interoperability standard including both products
- Denise and other users of overlay cells to investigate whether S-125 could be used as a buoy overlay

Interoperability and S-98, action points (2)

- COMEX² to check if S-102 and S-104 could be used instead of S-402 and the standardized waterlevel XML (→ own agenda point)
- COMEX² to investigate whether it is necessary to have an interoperability standard for areas with S-101 and S-401

Development of S-401

IEHG has to develop

- S-401 Product Specification
- S-401 Data Classification and Encoding Guide (DCEG)
- S-401 Feature Catalogue
- S-401 Portrayal Catalogue
- Conversion Guidance
- S-402 Product Specification for bathymetric IENCs (vector):
  development will not start before finalization of draft S-401
- Maybe Product Specifications, FCs and PCs for overlay IENCs and basic IENCs for use with overlays
- S-158:401 Recommended Validation Checks for IENCs
  including test cells
- S-403 inland interoperability standard

Github repository

- A **GitHub** repository (https://github.com/iehg) has been created
- The repository is used to share all documents and items which are part of the standard
- Possibility to **share issues**
- Seven repositories:
  - Product Specification
  - Feature Catalogue
  - Portrayal Catalogue
  - DCEG
  - Conversion Guidance
  - Validation Checks
  - Test cells

S-401 Product Specification

- IEHG has adopted edition 1.2.0 of the main document (based on edition 2.0.0 of S-101) last year
- The correction of some errors in the FC required a new edition 1.3.0 of S-401 which has also been adopted and published
- A draft of an edition 2.1.0 of S-101 has been published in July 2026
  but will still be amended before publication (expected for end of 2026)
- Corrections and amendments of this draft edition have been copied to the draft of edition 2.0.0 of S-401 and are marked as changes
- Document on the GitHub: https://github.com/IEHG/Product-Specification

Action points from last meeting (PS)

- Bernd to do an editorial check, finalize the document with the support of Claudia (regarding figures) and to publish a redline version and a clean version of edition 1.2.0 on the website
- All regions to test edition 1.2.0 of S-401 and to submit feedback
- COMEX² to try to include overlay cells in the next edition of S-401 and to report to IEHG (would have to be handled by S-98 or S-403 if separate PS for overlay cells; To be discussed whether it is possible and necessary to include them in S-401)
- Bernd and Gert to check the corrections of S-101 when they are available and to integrate the applicable ones in the draft of edition 2.0.0

S-401 main document: open question 1

- Under 8.5 we mentioned “S-403 Data Product Interoperability in S-100 Navigation Systems”
- The title of S-98 is changed to “S-100 ECDIS and Interoperability Specification”
- Which title will we use for S-403?

  “S-100 based Inland ECDIS and ECS Interoperability Specification”?

S-401 main document: open question 2

- Under 11.3.1 of S-101 the following sentence will be added:
  “When a dataset is created it must be supported with a digital signature which authenticates it against the Producer’s public key included in the Exchange Set metadata (see clauses 11.2 and 12.1.1). S-100 Part 15 defines the algorithms for compressing, encrypting and digitally signing datasets based on the S-100 Data Model.”
- Should this requirement also be added in S-401?
- Will S-401 IENCs only be signed, or also encrypted?

S-401 main document: open question 3

- The table under 12.1.3.1 of S-101 will contain the following sentence in the line for TIFF: “This is the only format allowed for a pictorial ENC Support file.”
- We have decided to allow also JPEG2000 in S-401
- Do we simply delete the sentence or do we replace it by something else?

S-401 Product Specification – way forward

- Final draft of S-401 edition 2.0.0 (based on S-101 edition 2.1.0) for IEHG 2027
- Open questions?
- Who is able to contribute?
- Time line?

- Action points:

Development of the S-401 DCEG

- The S-401 DCEG edition 1.2.0 has been developed based on the S-101 DCEG edition 2.0.0 and has been adopted by IEHG last year
- After the detection of some errors a corrected edition 1.3.0 of S-401 has been adopted and published.
- The corrections and amendments of the draft of edition 2.1.0 of S-101 have been copied (as far as relevant) to the draft of edition 2.0.0 of S-401 and are marked as track changes
- Some inland specific errors that have been detected in the meantime and the adopted change requests have also been added in track changes
- Document on the GitHub: https://github.com/IEHG/DCEG

Action points from last meeting DCEG

- Bernd to do an editorial check of the DCEG (including the replacement of the simple attribute SORIND with the new complex attribute) and provide the draft of edition 1.2.0 redline on GitHub
- Magdalina to check the associations in the DCEG and the FC for discrepancies and to report to Gaël, Gert and Bernd for correcting the FC (and/or the DCEG)
- Bernd to check that only “diesel fuel” is used in S-401 once the change has been accepted in the registry
- Bernd to create a clean version of the DCEG 1.2.0
- Bernd to publish the redline and the clean version on the website

DCEG: open question 1

On the pages for depth area the following text will be added in S-101:
“11.6.6 Subsurface cliffs not indicatable by contour interval or sounding pattern

Vessel performance and/or manoeuvrability may be impacted by very rapid, or even vertical, changes in the gradient of the sea floor. Vessels conducting sub-surface activities, including anchoring, may also be impacted by such rapid depth changes. In some cases, due to the contour interval and sounding pattern included on the chart, it is difficult if not impossible to adequately depict this information using depth contours and soundings. Where this is the case and these “subsurface cliffs” are considered to be important information for the boatmaster, these should be included on the largest optimum display scale IENC covering the area. If it is required to encode a subsurface cliff not indicatable by contour interval or sounding pattern, it must be done using the feature Slope Topline (see clause 5.11), with Boolean attribute in the water set to True.

Remarks:

DCEG: open question 1 continued

Remarks:

- The cliff top must be encoded in its true position. Sounding features should be included to represent the depths at the top and the bottom of the cliff; and may be repeated at intervals along the cliff if it is of significant extent at the optimum display scale for the data.
- This encoding must not be used to replace features considered to be obstructions to navigation; to supplement any charted obstructions to navigation; along the edges of dredged areas; or to indicate a steep drop-off from the coast. Steep drop-offs from the edges of obstructions to navigation, dredged areas or the coastline can be indicated by appropriate soundings selected adjacent to the obstruction, dredged area or the coastline.”

Does the new text of S-101 require inland specific clarifications?

DCEG: open question 2

S-101 introduces a new feature Depth discontinuity (A discontinuity between charted bathymetry resulting from time differences between hydrographic surveys in changeable areas or adjoining areas having significant differences in the quality of bathymetric data.)

Do we include this feature in S-401?

If yes, (M), (C) or (O)?

If yes, which SCAMIN?

If yes, are the associations Update Information, Additional Information and Spatial Association sufficient?
If yes, do we need inland specific Encoding Instructions?

Conversion guidance?

DCEG: open question 3

S-101 deletes enumeration 7 (ferry) from the attribute category of cable of the features Cable Submarine (CBLSUB) and Submarine Cable Area (CBLARE)

Submarine cables of ferries have to be encoded now as Ferry Route (FERYRT), if necessary combined with a Restricted Area (RESARE)

Do we follow S-101?

→ Conversion guidance

DCEG: open question 4

S-101 deletes enumeration 9 (anchorage for periods up to 24 hours) from the attribute category of anchorage (CATACH) of the features Anchorage Area (ACHARE) and Anchor Berth (ACHBRT) and replaces it by the new attribute maximum anchorage time

Do we follow S-101?

→ Conversion guidance

DCEG: open question 5

S-101 deletes enumeration 21 (dredging area) from the attribute category of restricted area (CATREA) of the feature Restricted Area (RESARE)

Dredging Areas have to be encoded now as Offshore Production Area with attribute category of offshore production area 5 (seabed material extraction area)

The extent and nature of any restricted area related to the extraction (or dredging) area must, where required, still be encoded using a Restricted Area feature (but with another enumeration)

Do we follow S-101?

→ Conversion guidance

DCEG: open question 6

S-101 deletes enumeration 18 (bearing light) from the attribute category of light (CATLIT) of the features Light All Around (LIGHTS) and Light Sectored (LIGHTS)

“Attribute category of light, value 18 (bearing light) included in S-101 Edition 2.0.0 has been removed in this Edition of S-101, as it is no longer required to be included on charts.”

Do we follow S-101?

→ Conversion guidance

S-401 DCEG and way forward

- Adoption of S-401 DCEG 2.0.0 (based on S-101 edition 2.1.0) by IEHG 2027
- Open questions?
- Who is able to contribute?
- Time line?

- Action points:

S-401 Feature Catalogue

- The S-401 FC edition 1.2.0 has been developed based on the S-101 FC edition 2.0.0 and has been adopted by IEHG last year
- After the detection of some errors a corrected edition 1.3.0 of S-401 has been adopted and published
- Still to do: integration of adopted change requests
- Open issues on the GitHub:
  - Issues · IEHG/Feature-Catalogue (github.com)

Action points from last meeting (FC)

- Gert to align the FC with the decisions of the meeting regarding the DCEG (including the replacement of SORIND with the new complex attribute)
- Bernd to publish the FC 1.2.0 on the website
- COMEX² to check that only “diesel fuel” is used in S-401 once the change has been accepted in the registry
- S-100 representatives to solve the inconsistencies in the registry [10 2e errors in the S 100 registry 2023 10 25.docx of the meeting 2023]

S-401 Feature Catalogue, way forward

- Draft of S-401 FC edition 2.0.0 (based on S-101 edition 2.1.0) for IEHG 2027
- Introduction of a complex feature notice mark for several physical notice marks on the same spot?
- Open questions?
- Who is able to contribute?
- Time line?

- Action points:

S-401 Portrayal Catalogue (Gert)

- LUA scripts are under development
  - Copy/pasted S-101 scripts
  - Edited S-101 scripts for S-401
  - Developed inland specific scripts

- How to test the individual scripts?

Portrayal Catalogue (Gert)

- Still to do:
  - Conversion of the CS
  - Solution for features on the same position

- Open issues on the GitHub:
  - Issues · IEHG/Portrayal-Catalogue (github.com)

Action points from last meeting, PC (1)

COMEX² to

- continue as proposed in the presentation when a stable PCB is available, or continue with the manual adaptation. Contribution of all members via the discussion forum
- develop test data based on the input from Laszlo and René
- amend the Conditional Symbologies on the basis of available CS (OpenCPN) or CS of other manufacturers. If no machine readable inland CS can be obtained, check if the maritime CS can be amended manually
- consider real world pictures and complex attributes as possible solutions for the display of multiple notice marks at the same spot
- create the XML for the PC (partly done but CS have still to be converted to LUA scripts, support of Hyacinth and maybe use of PC verify).
- ensure that top marks are displayed with colour (as in the current PresLib)

Action points from last meeting, PC (2)

- All members to inform Gert if they are able to contribute to the development of the Portrayal Catalogue.
- Tom, Friedhelm and Cameron to check if they can provide testing possibilities for LUA scripts
- All members that submit proposals with new symbols to provide the symbols in SVG format themselves or to provide them in a different format to Claudia Heckert for conversion to SVG. They have to take the new guideline and the form for symbols into account.
- Bernd to inform the IEHG members when it is clear whether the online meeting will take place

S-401 Portrayal Catalogue, way forward

- Draft of S-401 PC edition 2.0.0 (based on S-101 edition 2.1.0) for IEHG 2027
- Integration of adopted change requests
- Open questions?
- Who is able to contribute?
- Who will improve the LUA scripts if necessary?
- Who will provide a testing possibility?

- Time line?

Development of S-401 test cells

Action point from last meeting:

- Gert to forward the FC 1.2.0 to Caris and Esri together with the 2.5 test cells
- Tom and Gert to discuss conversion of the existing test cells following the meeting.

Development of S-401 test cells

- S-401 test cells edition 1.2.0
- Draft of S-401 test cells edition 2.0.0 for IEHG 2027
- Open questions?
- Who is able to contribute?

- Time line?

S-401 Alert catalogue

The last meeting came to the following conclusions:

- Europe: Inland ECDIS standard defines all kind of alarms, there's no need to focus on an 'alarm catalogue'
- US and Brazil: no immediate need

- The meeting could discuss whether they are still valid

S-158:401 Validation Checks

- The meeting could discuss the first draft of the S-158:401 Validation Checks and answer open questions

  Cover page
  Tables
  available on github

- The draft has to be updated in accordance with the decisions regarding the PS, DCEG and FC

S-57 to S-101 converter development

Action points from last meeting:

- Gert and Gaël to start working on the inland specific features
- Tom and Gert to start working on the copied and amended features (check all existing maritime rules). The first draft of S-101 to S-57 conversion guidance is available as S-65 Annex C
- Comex² to take the change from catach_10 (anchorage for pushing-navigation vessels) in IENC 2.5 to categoryOfAnchorage 16 (anchorage for pushing-navigation vessels) in S-401 into account for the Conversion Guidance
- COMEX² to take the changes regarding CONDTN 3, STATUS 18 and InDispute in CR 426 into account for the Conversion Guidance

Conversion guidance

- Draft
  available on github
- Stable version of the FC and DCEG are needed
- Integration of adopted change requests
- Is a conversion guidance S-401 to S-57 necessary?

S-401 conclusions and timeline

Action point from last meeting:

- The Core Group to develop a description of the options and pros and cons for the transitional period

The meeting could discuss proposals for a description of options for the transitional period

S-401 transitional period

- It will not be possible to switch from S-57 to S-401 on a specific date (replacement of all existing IENCs and all existing ECS/Inland ECDIS, at least within a region)
- We will therefore need a transitional period, in which either
  - the chart producers will provide S-57 based IENCs and S-401 IENCs or
  - the manufacturers of ECS/Inland ECDIS will provide dual fuel applications that can read S-57 based IENCs and S-401 IENCs
- IEHG decided in 2025 that decisions regarding the transitional period have to be taken on regional level
- IEHG has developed a description of options for the transitional period

S-401 transitional period: options

- hard cut for both sides: on a specific date all applications on vessels have to be exchanged and all IENCs have to be published as S-401 (not realistic)
- Dual fuel for both sides (like the maritime area): chart producers publish edition 2.6 IENCs AND S-401 IENCs, dual fuel applications can use both formats
- Dual fuel only for chart producers: chart producers provide 2.6 and S-401, users can use either old applications or new applications, but do not need dual fuel
- Dual fuel only for applications: all users have to use dual fuel applications, chart producers can gradually replace their 2.6 IENCs with S-401, but do not need to publish the same IENCs in both formats

S-401 transitional period, dates to be fixed

Some transitional periods are consisting of several periods that have to be defined

e.g. the most supported concept for Europe:

- From entry into force of S-401 until date 1 all vessels have to be equipped with dual fuel applications
- From date 1 until date 2 chart producers can either publish Edition 2.6 or S-401 Chart producers with many IENCs will replace their 2.6 IENCs with S-401 IENCs step by step, but date 2 is the deadline for a full coverage with S-401 IENCs
- From date 2 onwards only S-401 IENCs are published

S-401 conclusions and timeline

- Do we want to adopt S-401 edition 2.0.0 in spring or in autumn 2027?
- When will we be able to adopt S-403?
- Open questions?
- Who is able to contribute?

- Time line?

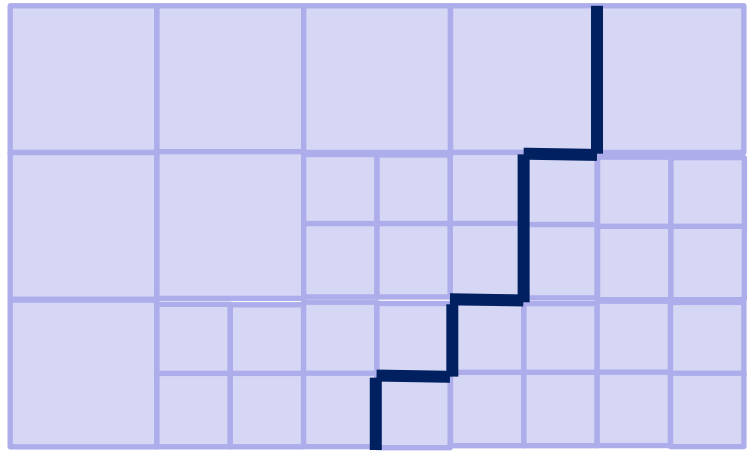
The Bathymetric Contour Overlay (S-402)

- S-402 has been reserved by IHO for the Bathymetric Contour Overlay for Inland ENC Product Specification as requested by IEHG. The existing Product Specification for bathymetric Inland ENCs and S-401 should be the basis for the development of S-402. IEHG has decided to start with the development of S-402 when the corresponding parts of S-401 have been adopted.

- But we need to decide for S-403 whether we will use the bathymetric contour overlay S-402 (in combination with water level information in XML-format) or S-102 (gridded depth data) and S-104 (gridded water level information)

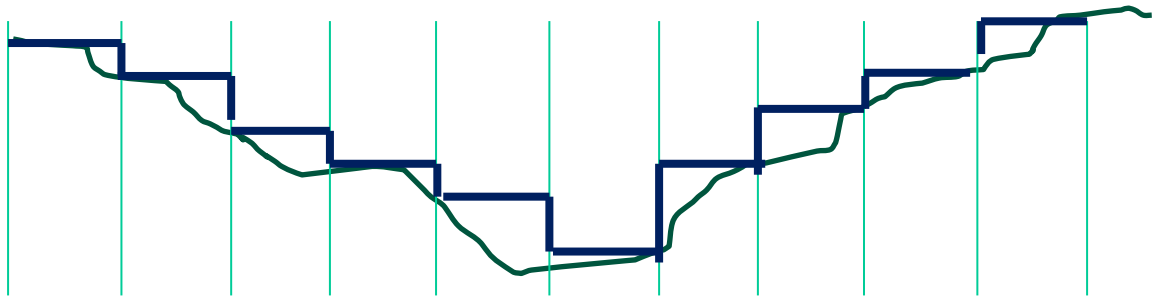
S-102 Bathymetric Surface (1)

- IHO recommends to consider the use of S-102 (Bathymetric Surface Product Specification) for inland waterways
- S-102 provides depth information as gridded raster data while S-101, S-401, bathymetric IENCs and S-402 provide depth areas and depth contours as vector data
- Grid size may vary depending on the required accuracy
- Contours are created on the fly by the ECDIS, they are rectangular and depending on the grid size



S-102 Bathymetric Surface (2)

- The depth value of each grid is the least sounded depth within this grid this leads to a loss of accuracy



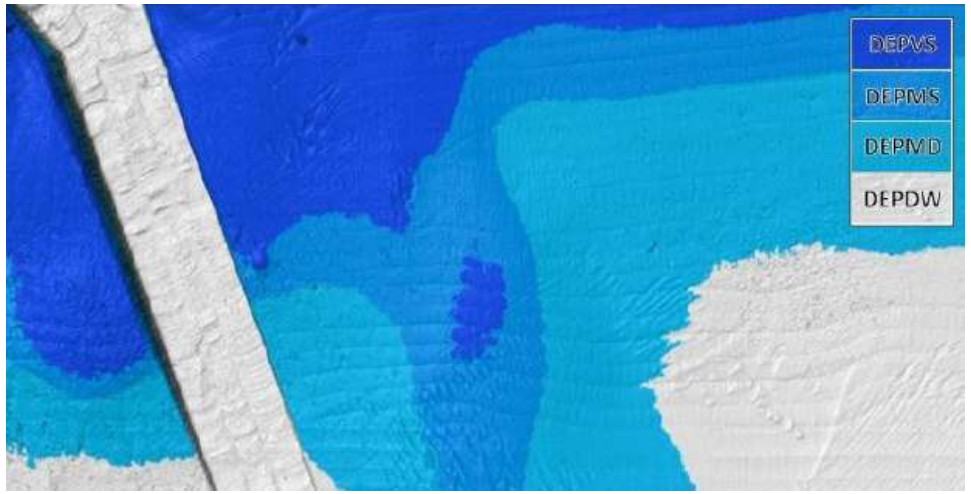
- The effect is similar to the effect of depth contours
- To reach the same accuracy as 10 cm depth contours with a 2 m grid the slope may not have more inclination than 1:20

S-102 Bathymetric Surface (3)

- At the moment S-102 does not mention the inland specific values of the attribute Vertical Datum (e.g. Equivalent height of water (German GlW), Highest Shipping Height of Water (German HSW), Reference low water level according to Danube Commission, Dutch river low water reference level (OLR))
- The missing values can be added in S-102, but the timeline is unsure
- Chart producing authorities that use depth data from single beam surveys still have to investigate how much effort is needed to create gridded data

S-104 water level information

- Maritime ECDIS is using S-102 depth data in combination with S-104 water level information
- S-104 provides the water level information in a rectangular grid
- The ECDIS has to calculate the difference between the two surfaces
- S-104 water level information can be produced from a 3-D waterlevel model
- The transformation of a 2-D model into S-104 data is difficult and inaccurate

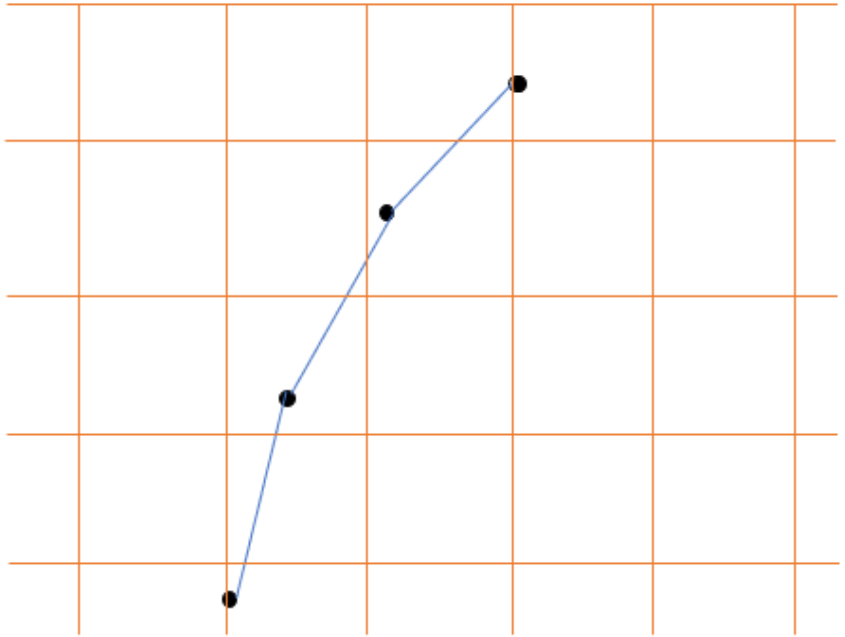


Edition 2.x water level information

- Each depth area in an IENC for the use with a water level model is containing the attribute waterway distance
- The standardized water level XML is providing a depth correction value for each waterway distance value
- The IECDIS is simply adding the correction value to the depth value in the IENC
- Example:
  - at river-km 123,400 there are depth areas with
    1,0 m – 1,1 m – 1,2 m – 1,4 m - …
  - the correction value for 123,400 is +0,2 m
  - The IECDIS will display the depth areas with
    1,2 m – 1,3 m – 1,4 m – 1,6 m - …

Creation of an S-104

- 2-dimensional water level models are just providing water level information for each waterway distance value
- This point information has to be converted to gridded data for S-104
- This will again decrease the accuracy of the Information



Description of the options

1. S-102 (gridded depth data) and S-104 (gridded water level data)
   Pro: compatible with maritime
   Con: decreased accuracy, unsure timeline (missing vertical datums)

2. S-402 (bathymetric IENCs with depth areas and depth contours) and standardized water level XML as a support file
   Pro: high accuracy, easy production
   Con: not compatible with maritime

3. S-402 (bathymetric IENCs with depth areas and depth contours) and S-40x (Product Specification for a S-100 based water level information for inland waterways)
   Pro: high accuracy, easy production, technically compatible with maritime
   Con: high development effort

S-201 and S-125

- IALA is working on Product Specifications for Aids to Navigation (AtoN) Information (S-201) and for Navigation Services (S-125). The meeting could be informed about the status of development

Annual report to HSSC

Action point from last meeting:

- Denise and Bernd to prepare a report for the next HSSC taking into account the developments regarding S-98
  Status: Done, action point closed

New report for the next meeting:
Who, when?

Information document on Inland ENCs

- IEHG has developed a document that provides information on Inland ENCs and the relation and the differences to maritime ENCs

- The document
(available on the ienc.openecis.org website)
can be used by every member of IEHG
for presentations, discussions, explanations

- All members of IEHG are invited to submit proposals for amendments of the information document on Inland ENCs

Information (EN)

Información (SP)

Información (PT)

Information (DE)

Informacion (FR)

Informacion (NL)

Information document on Inland ENCs

Action points from last meeting:

- All members of IEHG are invited to submit proposals for amendments of the information document on Inland ENCs (Inland ENC.ppt on the IEHG website)

- The Core Group to update regulatory status, coverage and contact persons

Operation of Web Services (1)

Future operation of the ienc.openecdis.org website

- now hosted by the European Committee for drawing up standards in the field of inland navigation (CESNI)

- Proposals for improvements?

https://ienc.openecdis.org

Operation of Web Services (2)

Discussion forum

- https://github.com/orgs/IEHG/discussions
- Please contribute!

Operation of Web Services (3)

- IENC domain in the registry of IHO

http://registry.iho.int

Any other business

- How to encourage other countries to join IEHG?
  South America / Asia
- Any other business?
- Departures

Next Meeting

- Should we have a 2 hour online meeting once a month until we have finalized S-401 edition 2.0.0?
- Do we need a physical meeting in 2027or is an online meeting sufficient?
- where and when ?
  Last meetings: 2015 China, 2017 Brazil, 2019 US, 2021 Europe, 2022 online, 2023 US, 2024 China, 2025 Europe
- Denise had offered to host the IEHG meeting 2026 in Louisville
- Changes to the agenda of next meeting?

Thank you all for participating!

Have a good night/day!